

SEISMOGRAPH BASED ON ULTRASONIC, HALL-EFFECT AND PIEZOELECTRIC SENSORS

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I. AIM

To detect earthquakes and warn the users based on their relative intensity. The device should also be able to plot the graph of the intensity of vibrations so that it can be a low-cost substitute for a conventional seismograph.

The device should sound an alarm in case of an earthquake. For minor earthquakes, it should light up one LED; for medium intensity, two LEDs; for major earthquakes, it should light up all three LEDs. On connecting the device to the IDE in the computer, it should plot the magnitude of the seismic waves like a seismograph against time.

II. CONCEPTS USED FOR APPLICATION

- Use of an Arduino
- Interface of Arduino with various sensors
- Working of a Piezoelectric sensor
- Working of an Ultrasonic sensor
- Working of a Hall effect sensor
- Interface of Arduino with LEDs and buzzer
- Calibration of the output with a standard scale
- Plotting the output using Arduino IDE

III. THEORY AND WORKING PRINCIPLE

An Arduino is a microcontroller board that we are using as an interface between the sensors and the output devices, such as LEDs, buzzer, and serial plotter in the Arduino IDE. A Hall effect sensor is a sensor that is used to measure the magnitude of the magnetic field at a place. A piezo-electric sensor is a sensor that is used to measure changes in mechanical stress using the piezoelectric effect. An ultrasonic distance sensor is a sensor that measures its distance from another point by using ultrasonic sound waves. [1]

This device consists of a bar magnet suspended from the ceiling of an enclosure with a thread. The seismic vibrations will cause this bar magnet to vibrate along its mean position. One of the ends of this thread is connected to one of the plates of the piezoelectric sensor. Any vibration due to the earthquake will induce some vibrations in the bob (bar magnet) and a change in tension in the thread, which will result in a change in the stress across the piezoelectric sensor. This will induce the creation of charge across the plates, which the Arduino board will detect and, therefore, ring the buzzer. We also use an Ultrasonic sensor and a Hall effect sensor in the device to monitor the bob's motion in case of an earthquake. The Ultrasonic distance measuring sensor will measure the deflection of the magnet from its mean position in one axis, while the Hall effect sensor will measure the deflection of the angular as well as Euclidean position of the bar magnet from its mean position calculated by the change in the magnetic field. The sum of the absolute scaled values of the deflection of the bob from the equilibrium position as measured from the two "strategically" placed sensors will be plotted in the "serial plotter" of the Arduino IDE. Finally, we included LED bulbs in the circuit to alarm the earthquakes according to their relative intensities as measured by the magnitude of deflection from the mean position. Weaker earthquakes will light up only one LED, while medium and high-intensity earthquakes will light up two and three LEDs, respectively. This will be based on the amount of deflection to cross a certain threshold value to activate the LEDs and buzzer.

A. Components Used

TABLE I. LIST OF COMPONENTS USED

Components Used	Quantity
Arduino Uno R3	1
Piezo Buzzer	1
Piezoelectric Sensor	1
Breadboard	1
Power adapter and cable	1
Ultrasonic Distance Measuring Sensor	1
Hall effect sensor	1
LEDs	3
Connecting wires	As required

B. Structure of the Device (Procedure)

Fig. 1. Final model of the seismograph



Fig. 2. Positioning of the Sensors

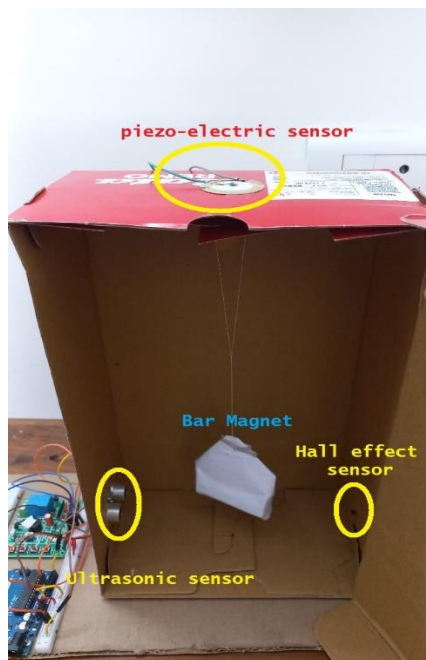


Fig. 4. Plots of the seismic waves as detected by the seismograph on Arduino IDE

C. Circuit Connections (Procedure)

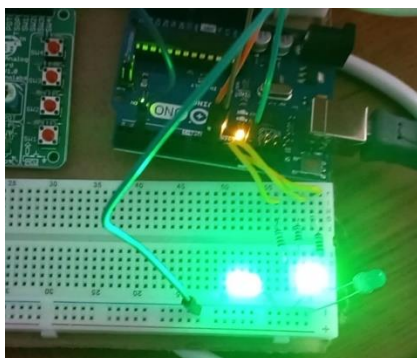
The output LEDs and buzzer are connected to the digital pins of the Arduino board. The Piezoelectric and Hall effect sensors are connected to the analog pins of the Arduino board. The Ultrasonic sensor is connected to digital PWM-enabled pins of the Arduino board.

D. Structure of the Code on Arduino IDE (Procedure)

The code first contains the setup block, which contains the instructions for setting up various pins as input and output. Then, it contains the loop block in which the outputs from the sensors were calibrated and added to give the total deflection of the bar magnet from its mean position due to the earthquake. Then, certain 'if-else' blocks were created, which contained the instruction of switching ON the LEDs and buzzer when the earthquake's magnitude crossed a certain threshold. There was also a block of code meant to serial plot the magnitude of deflection in Arduino IDE.

IV. RESULTS

Fig. 3. Earthquake intensity as indicated by the LEDs



SIGNIFICANCE OF THIS PROJECT

The idea of detecting, measuring and plotting the magnitude of an earthquake using distance-measuring ultrasonic sensor, hall effect sensor, piezoelectric sensor and bar magnet is very unique idea of creating seismographs. There exist some seismographs which use sophisticated and expensive sensors like 3- axis accelerometers and sensitive gyroscopes. They have a very high cost, as high as a few thousand rupees. Our device uses sensors, which have a total cost of less than a hundred rupees. The sensors should be accurate and cheaper as well. So, we are using three sensors with much less cost, which will monitor the motion of the suspended bar magnet in the 3 perpendicular axes. This seismograph can plot as well as alarm about the seismic waves. So, we can use this device for a commercial as well as scientific purposes also.

Therefore, we use this UNIQUE idea to form "cheap and decently accurate" seismographs for large-scale deployment for a more detailed study of earthquakes as well as save many lives from earthquakes.

V. REFERENCES

- [1] L. Reese, "The working principle, applications and limitations of ultrasonic sensors," *Microcontroller Tips*, 2019. [Online]. Available: <https://www.microcontrollertips.com/principle-applications-limitations-ultrasonic-sensors-faq/>. [Accessed: 24-Apr-2023].